Predicting Arrest Outcomes in Terry Stops

A Data-Driven Analysis of Police Conduct and Racial Equity for the Seattle City Council Public Safety Committee

Business Understanding & Stakeholder Goals

Analyzing the predictors of arrest to ensure accountability and fairness in Seattle's public safety operation

Overview of Arrest Frequency

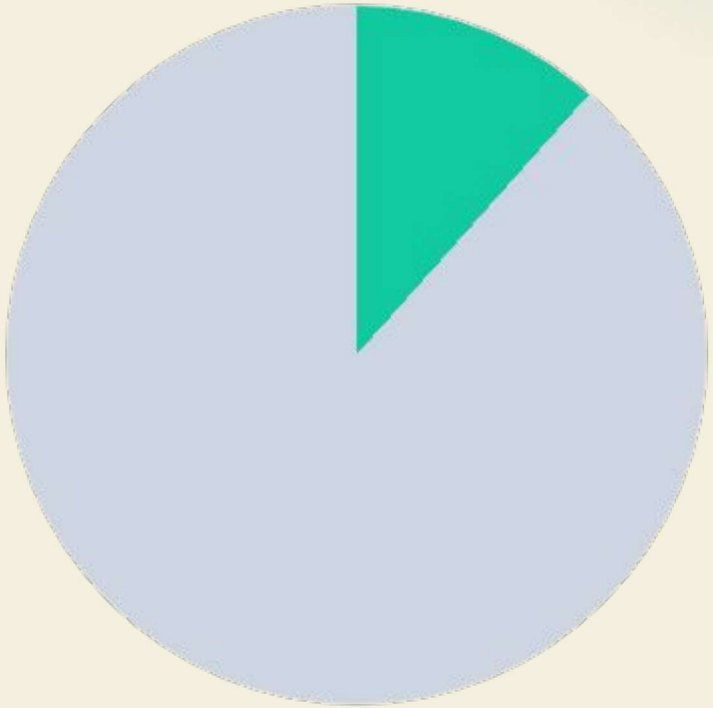
Scale of the Dataset

The analysis is based on **67,362 recorded incidents** of Terry Stops. A key characteristic is the heavy "class imbalance":

Arrest: 59,542 (88.4%)

rest: 7,820 (11.6%)

Most stops do not lead to an arrest, making the identification of "Arrest Predictors" statistically critical.



Subject Demographics Profile

Gender

Terry Stops are heavily skewed by gender, with **Male subjects** accounting for 53,119 incidents compared to 13,449 for Females.

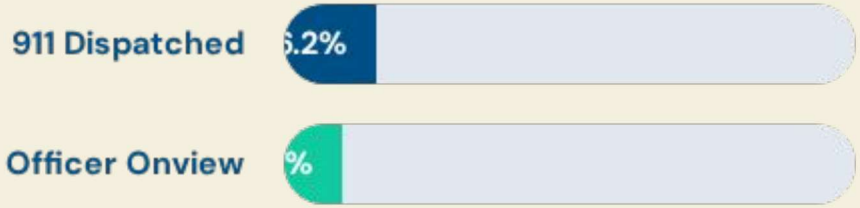
Age Distribution

The **26-35 age group** is the most frequent (22,385), followed by 36-45. Stops are concentrated in working-age adults.

Perceived Race

White (32,859) and Black African American (20,40 subjects represent the va majority of all police-civi contacts.

Initiation Method & Outcomes



Predictive Circumstances

Stops initiated via **911 emergency calls** result in a 60% more often than stops initiated directly by o view.

This highlights the importance of the initial context—reported crimes provide stronger "proba cause" than proactive suspicion alone.

Graphic representing racial equity and policing metrics

Investigating Disparities

Arrest Rates by Race

Raw data shows significant variation in arrest probability across perceived racial groups:

- **Native Hawaiian/Islander:** 24% arrest rate
- **Black or African American:** 13% arrest rate
- **White:** 11% arrest rate

These numbers alone don't prove bias, but they serve as the "Critical Question" for our machine learning audit.

Machine Learning Methodology

Moving beyond raw numbers to find statistically meaningful signals.

Comparing Classification Models

Model Tested	Accuracy	Recall (Arrest Class)	F1 Score
Logistic Regression	88%	0%	0.00
Decision Tree	88%	3%	0.06
Random Forest (Best)	57%	78%	0.30

*Note: While Random Forest has lower accuracy, it is the **most reliable** because it actually identifies arrests (78% re*

whereas others simply guess "No Arrest."

The Drivers of Arrest

The model reveals that **Race** is a meaningful predictor of arrest even when holding circumstantial factors (like weap
types) constant. It ranks higher than gender or frisk status in predictive weight.

Evaluating Racial Predictive Weight

A Non-Neutral Finding

The model found that knowing the **perceived race of the subject** and the **race of the officer** genuinely improved its ability to predict an arrest.

In a purely objective system, race should carry **zero** predictive weight. The fact that it persists as a signal suggests that demographics play a statistically significant role in policing outcomes in Seattle.

Strategic Recommendations

- **Audit Officer Discretion:** Conduct deeper reviews into "Onview" stops, where arrest rates are lower but the po
subjective bias is higher.
- **Standardize Reporting:** Improve data collection for "Weapon Type," which was missing for over 93% of records
the model's performance.
- **Focused Equity Audits:** Use the model's findings to target precincts or units where race shows the highest pr
weight for arrest outcomes.